```java
import java.util.ArrayList;
import java.util.Scanner;

public class StudentGradeTracker {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        ArrayList<String> studentNames = new ArrayList<>();
        ArrayList<Integer> studentGrades = new ArrayList<>();

        System.out.print("Enter number of students: ");
        int numStudents = scanner.nextInt();
        scanner.nextLine(); // Consume newline

        for (int i = 0; i < numStudents; i++) {
            System.out.print("Enter student name: ");
            String name = scanner.nextLine();
            studentNames.add(name);

            System.out.print("Enter grade for " + name + ": ");
            int grade = scanner.nextInt();
            scanner.nextLine(); // Consume newline
            studentGrades.add(grade);
        }

        // Calculate average, highest, and lowest
        int total = 0;
        int highest = Integer.MIN_VALUE;
        int lowest = Integer.MAX_VALUE;

        for (int grade : studentGrades) {
            total += grade;
            if (grade > highest) highest = grade;
            if (grade < lowest) lowest = grade;
        }

        double average = (double) total / studentGrades.size();

        // Display results
        System.out.println("\n--- Grade Summary ---");
        for (int i = 0; i < studentNames.size(); i++) {
            System.out.println(studentNames.get(i) + ": " + studentGrades.get(i));
        }

        System.out.println("Average Grade: " + average);
        System.out.println("Highest Grade: " + highest);
        System.out.println("Lowest Grade: " + lowest);

        scanner.close();
    }
}
```

```
Enter number of students: 2
Enter student name: Abi
Enter grade for Abi: 50
Enter student name: Anu
Enter grade for Anu: 51

--- Grade Summary ---
Abi: 50
Anu: 51
Average Grade: 50.5
Highest Grade: 51
Lowest Grade: 50

=== Code Execution Successful ===
```

```java
import java.util.HashMap;
import java.util.Scanner;

public class StockTradingPlatform {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Simulated stock prices
        HashMap<String, Double> market = new HashMap<>();
        market.put("AAPL", 150.0);
        market.put("GOOGL", 2800.0);
        market.put("AMZN", 3400.0);
        market.put("TSLA", 700.0);

        // Portfolio and balance
        HashMap<String, Integer> portfolio = new HashMap<>();
        double balance = 10000.0; // Starting amount

        boolean running = true;

        while (running) {
            System.out.println("\n--- Stock Trading Platform ---");
            System.out.println("1. View Market Data");
            System.out.println("2. Buy Stock");
            System.out.println("3. Sell Stock");
            System.out.println("4. View Portfolio");
            System.out.println("5. Exit");
            System.out.print("Choose an option: ");
            int choice = scanner.nextInt();
            scanner.nextLine(); // consume newline

            switch (choice) {
                case 1:
                    System.out.println("\n--- Market Data ---");
                    for (String symbol : market.keySet()) {
                        System.out.println(symbol + ": $" + market.get(symbol));
                    }
                    break;

                case 2:
                    System.out.print("Enter stock symbol to buy: ");
                    String buySymbol = scanner.nextLine().toUpperCase();
                    if (market.containsKey(buySymbol)) {
                        System.out.print("Enter quantity to buy: ");
                        int qty = scanner.nextInt();
                        double cost = qty * market.get(buySymbol);
                        if (cost <= balance) {
                            balance -= cost;
                            portfolio.put(buySymbol, portfolio.getOrDefault(buySymbol, 0) + qty);
                            System.out.println("Bought " + qty + " shares of " + buySymbol);
                        } else {
```

```
--- Stock Trading Platform ---
1. View Market Data
2. Buy Stock
3. Sell Stock
4. View Portfolio
5. Exit
Choose an option: 1

--- Market Data ---
GOOGL: $2800.0
AAPL: $150.0
TSLA: $700.0
AMZN: $3400.0

--- Stock Trading Platform ---
1. View Market Data
2. Buy Stock
3. Sell Stock
4. View Portfolio
5. Exit
Choose an option: 2
Enter stock symbol to buy: 
```

```java
import java.util.*;

public class HotelReservationSystem {
    static ArrayList<Room> rooms = new ArrayList<>();
    static ArrayList<Reservation> reservations = new ArrayList<>();
    static Scanner scanner = new Scanner(System.in);
    static double userBalance = 10000.0;

    public static void main(String[] args) {
        initializeRooms();

        boolean running = true;
        while (running) {
            System.out.println("\n--- Hotel Reservation System ---");
            System.out.println("1. View Available Rooms");
            System.out.println("2. Make Reservation");
            System.out.println("3. View Booking Details");
            System.out.println("4. Exit");
            System.out.print("Choose an option: ");

            int choice = scanner.nextInt();
            scanner.nextLine(); // consume newline

            switch (choice) {
                case 1 -> viewAvailableRooms();
                case 2 -> makeReservation();
                case 3 -> viewBookings();
                case 4 -> {
                    running = false;
                    System.out.println("Thank you for using the system!");
                }
                default -> System.out.println("Invalid option.");
            }
        }

        scanner.close();
    }

    static void initializeRooms() {
        rooms.add(new Room(101, "Standard", 3000.0));
        rooms.add(new Room(102, "Standard", 3000.0));
        rooms.add(new Room(201, "Deluxe", 5000.0));
        rooms.add(new Room(202, "Deluxe", 5000.0));
    }

    static void viewAvailableRooms() {
        System.out.println("\n--- Available Rooms ---");
        boolean anyAvailable = false;
        for (Room room : rooms) {
            if (!room.isBooked) {
                System.out.println("Room " + room.roomNumber + " - " + room.category + " - ₹" + room.price);
                anyAvailable = true;
            }
        }
        if (!anyAvailable) {
            System.out.println("No rooms available at the moment.");
        }
    }
```

```
--- Hotel Reservation System ---
1. View Available Rooms
2. Make Reservation
3. View Booking Details
4. Exit
Choose an option: 1

--- Available Rooms ---
Room 101 - Standard - ?3000.0
Room 102 - Standard - ?3000.0
Room 201 - Deluxe - ?5000.0
Room 202 - Deluxe - ?5000.0

--- Hotel Reservation System ---
1. View Available Rooms
2. Make Reservation
3. View Booking Details
4. Exit
Choose an option: 4
Thank you for using the system!

=== Code Execution Successful ===
```